Experiment 5 : Data Visualization using Matplotlib

Aim: Data Visualization using Matplotlib. Analyzing Air Quality Index (AQI) trends in a city datas

Objective: The dataset contains air quality measurements of a city over a period of time, with at such as date, pollutant levels (PM 2.5, PM 10, CO) and AQI values. The goal is to use the r library to create visualizations that effectively represent the AQI trends and patterns for different pollutants in the city.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
# Step 1: Import the City Air Quality dataset
df = pd.read_csv("aqi.csv", parse_dates=["date"])
print("Shape:", df.shape)
df.head()
```

```
Shape: (90, 4)
```

Out[2]:

	date	pm25	pm10	
0	2026-01-01	81.264043	121.701105	1.288900
1	2026-01-02	92.498783	126.602617	1.621444
2	2026-01-03	55.102277	127.325972	0.946455
3	2026-01-04	97.339963	106.894428	1.051758
4	2026-01-05	69.542462	153.414451	1.014588

In [3]:
```
# Step 2: Explore the structure and content of the dataset
df.info()
print()
print("Missing values:", df.isnull().sum().sum())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   date    90 non-null     datetime64[us]
 1   pm25    90 non-null     float64
 2   pm10    90 non-null     float64
 3   co      90 non-null     float64
dtypes: datetime64[us](1), float64(3)
memory usage: 2.9 KB

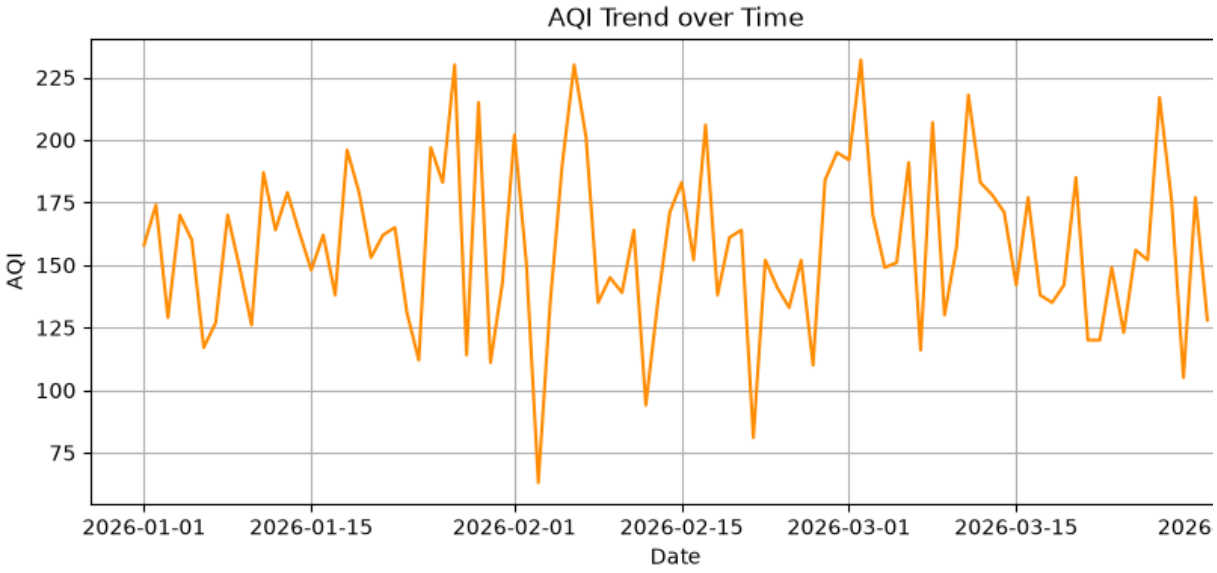
Missing values: 0
```

In [4]:
```
# Step 3: Identify relevant variables and calculate the AQI value
df["aqi"] = (df["pm25"] * 1.2 + df["pm10"] * 0.5).astype(int)
print("AQI minimum:", df["aqi"].min())
print("AQI maximum:", df["aqi"].max())
print(df.describe())
```

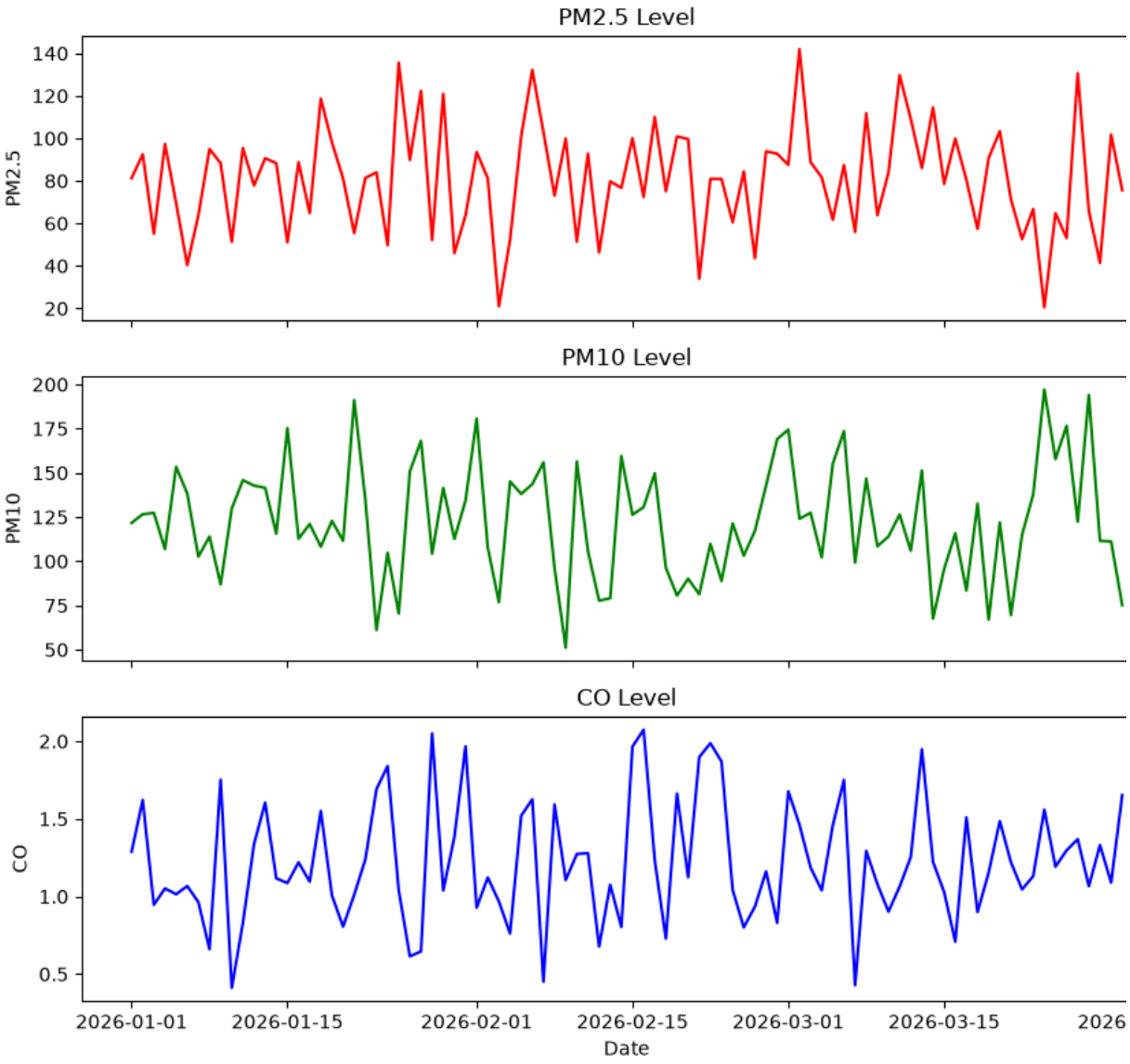
```
AQI minimum: 63
AQI maximum: 232
```

	date	pm25	pm10	co	aqi
count	90	90.000000	90.000000	90.000000	90.000000
mean	2026-02-14 12:00:00	80.918911	122.458523	1.221775	157.844444
min	2026-01-01 00:00:00	20.505618	51.189485	0.411299	63.000000
25%	2026-01-23 06:00:00	63.741348	103.431085	1.007870	135.000000
50%	2026-02-14 12:00:00	81.312431	121.526669	1.142707	156.500000
75%	2026-03-08 18:00:00	96.871658	143.381380	1.503508	179.000000
max	2026-03-31 00:00:00	141.914604	197.124741	2.074339	232.000000
std	NaN	25.418651	32.207356	0.388954	33.402520

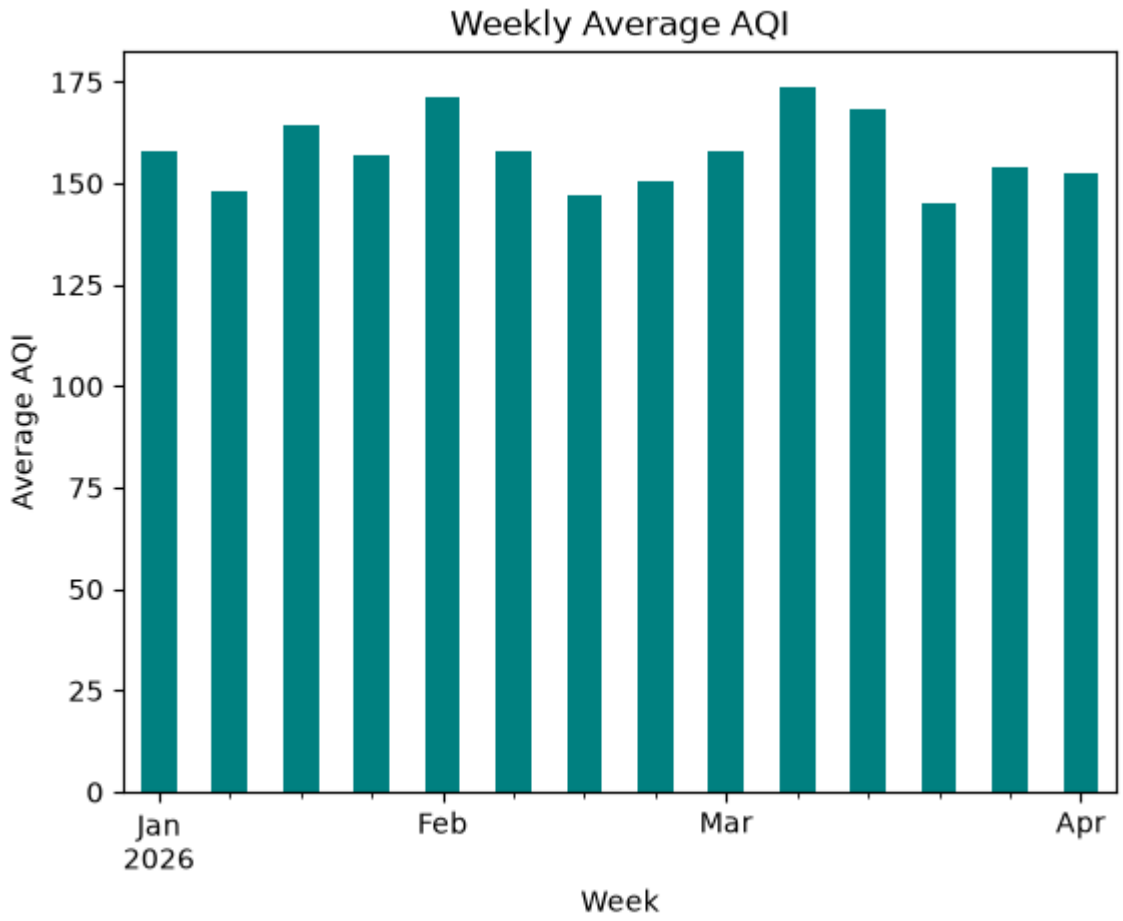
In [5]:
```python
# Step 4: Line plot of the overall AQI trend over time
plt.figure(figsize=(10, 4))
plt.plot(df["date"], df["aqi"], color="darkorange")
plt.title("AQI Trend over Time")
plt.xlabel("Date")
plt.ylabel("AQI")
plt.grid(True)
plt.show()
```



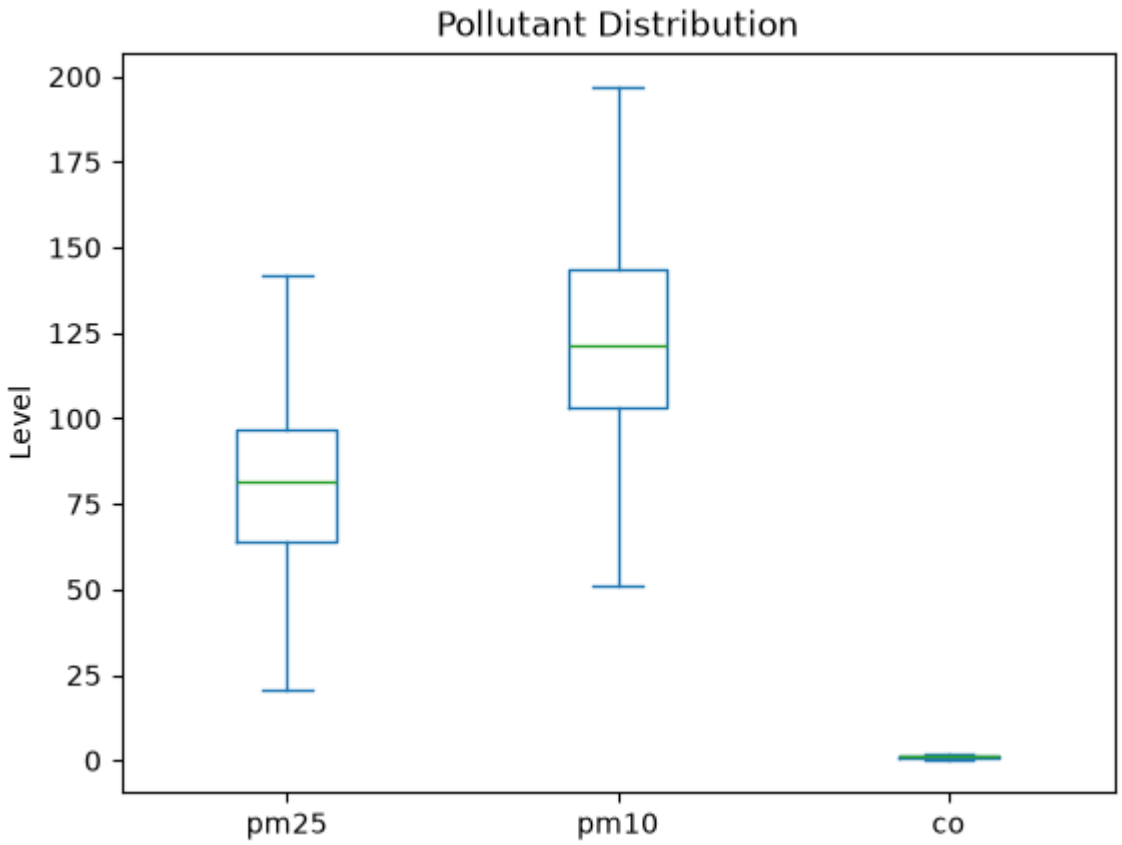
In [6]:
```python
# Step 5: Individual pollutant levels on separate line plots
fig, ax = plt.subplots(3, 1, figsize=(9, 8), sharex=True)
ax[0].plot(df["date"], df["pm25"], color="red")
ax[0].set_title("PM2.5 Level")
ax[0].set_ylabel("PM2.5")
ax[1].plot(df["date"], df["pm10"], color="green")
ax[1].set_title("PM10 Level")
ax[1].set_ylabel("PM10")
ax[2].plot(df["date"], df["co"], color="blue")
ax[2].set_title("CO Level")
ax[2].set_ylabel("CO")
ax[2].set_xlabel("Date")
plt.tight_layout()
plt.show()
```



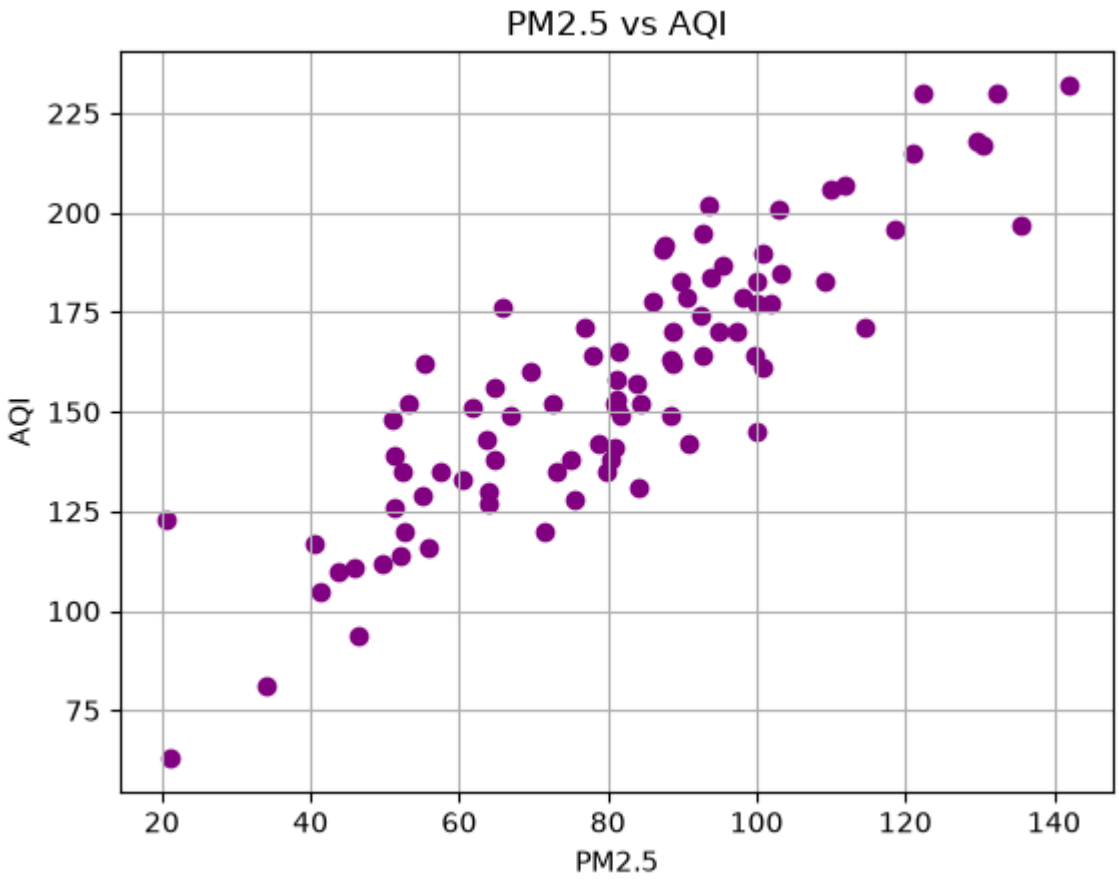
In [7]:
```
# Step 6: Bar plot of weekly average AQI
weekly = df.set_index("date")["aqi"].resample("W").mean()
weekly.plot(kind="bar", color="teal")
plt.title("Weekly Average AQI")
plt.xlabel("Week")
plt.ylabel("Average AQI")
plt.show()
```



In [8]:
```
# Step 7: Box plot of pollutant distribution
df[["pm25", "pm10", "co"]].plot(kind="box")
plt.title("Pollutant Distribution")
plt.ylabel("Level")
plt.show()
```



In [9]:
```
# Step 8: Scatter plot of PM2.5 against AQI
plt.scatter(df["pm25"], df["aqi"], color="purple")
plt.title("PM2.5 vs AQI")
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.grid(True)
plt.show()
```



Conclusion: Matplotlib is a powerful and flexible tool for data visualization in Python. Using it, th trends and pollutant patterns of the city were visualized through line plots, bar plots, box plots ar plots, turning raw air quality data into meaningful visual insights.